Characteristic curve of a vacuum diode and the Frank-Hertz experiment with neon

Luca Reisert and Stephan Hollemann

April 22, 2026

Electrons with a certain kinetic energy collided with neon atoms. At specific energies, a decrease in the anode current and a glow in the experimental setup were observed. The minima of the anode current demonstrated that atoms have discrete energy levels, and the excitation energy between two states could be calculated.
Furthermore, the characteristic curve of a vacuum diode was plotted and various quantities were derived from it. This aids in understanding the technical component. The anode current was investigated as a function of the accelerating voltage at different heating voltages. The results show that at higher temperatures, more electrons are emitted, resulting in a higher anode current at the same accelerating voltage. The Schottky-Langmuir space charge law could not be confirmed by these measurements. The contact voltage was calculated and the work function of the filament was determined. Finally, the saturation current was measured as a function of temperature.

1 Characteristic curve of a vacuum diode

1.1 Introduction

The experiment confirmed the postulated discrete energy levels of Bohr's atomic model. In this experiment, fundamental principles of a vacuum diode are determined. This is important to understand the quantom mechanical nature of atoms and thermoionic emission. For this purpose, the characteristic curve is plotted, the work function W_A is determined, the contact voltage U_K is measured, and the saturation current I_S is recorded at various heating voltages U_H. With the results, the behavior of electrons in a vacuum diode can be determined. Based on these findings, further experiments such as the Frank-Hertz experiment can be carried out. In this experiment, thermal energy E_{th} is supplied to valence electrons so that they overcome the work function W_A of a metallic material. The higher the temperature of the metal, the greater the number of emitted electrons. The Richardson equation [1] can be used to describe the process of electron emission from a metal.

$$I_S = B * T^2 * exp(-\frac{W_A}{k_B * T}) \tag{1}$$

For a correct description, the contact voltage U_K must be taken into account. The contact voltage is the potential difference arising from the different work functions of the electrode materials, which acts as an additional voltage between the cathode and anode. The electrons emitted from the metal collect in front of the positively charged metal and repel each other. An electron cloud is formed. An electron can be attracted to the anode by applying a voltage. The stronger the voltage, the further the electric field extends into the electron cloud. The stronger the electric field, the greater the current measured at the anode. This proportional relationship is described by the Schottky-Langmuir space charge law [2].

$$I \propto U_a^{\frac{3}{2}} \tag{2}$$

Above a certain voltage, the electric field penetrates the entire electron cloud, and all electrons are attracted directly to the anode. If the voltage is increased further, no more electrons can be attracted. A saturation current I_S is established.

1.2 Materials and methods

The vacuum diode consists of a tungsten filament, which serves as the cathode, and an anode. The electrodes are located in an evacuated chamber. An adjustable heating voltage is applied to the filament. This voltage causes the emission of electrons from the filament. A variable electric field is applied between the anode and cathode. This field is set by the accelerating voltage U_a. The anode current is measured at the anode as a function of the anode voltage using an ammeter. The characteristic curve is recorded as the anode current as a function of the

accelerating voltage at a set heating voltage. The experimental setup is shown in figure [1].

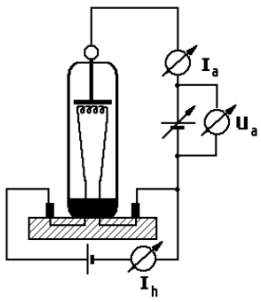

Figure 1: Schematic setup for determining the characteristic curve of a vacuum diode at different heating voltages.

1.3 Results and discussion

A characteristic curve was recorded for each of the following heating voltages $U_H = 6.5V$ and $U_H = 7.5V$. Acceleration voltages U_a were set in the range of $U_{a,min} = 0V$ to $U_{a,max} = 500V$. The step size in which the anode current was measured is $\Delta U_a = 10V$. The characteristic curves are shown in figure [2].

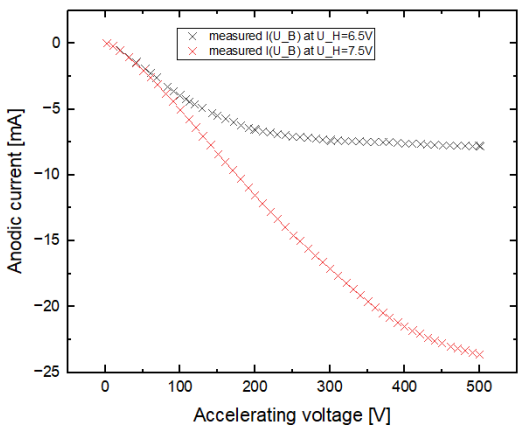

Figure 2: Characteristic curves of a vacuum diode recorded at the heater voltages $U_H = 6.5V$ and $U_H = 7.5V$. Measurements were taken from $U_a = 0V$ to $U_a = 500V$ in $\Delta U_a = 10V$ steps.

Both characteristic curves can be divided into three areas. Even at low voltages, a small anode current can be measured. This results mainly from the electrons, which, due to the thermal energy, have enough kinetic energy to reach the anode without an electric field being applied. As the voltage increases, the characteristic curve takes the form of a straight line. In this area, the electron cloud is dissipated in front of the cathode. Above a certain voltage U_a, the line flattens out and remains at a constant value. This value is called the saturation current I_S. The electron cloud was completely dissipated and all emitted electrons are collected directly by the anode. The saturation current depends on the number of emitted electrons. The more electrons are removed from the wire, the more electrons are collected. This is reflected in a larger saturation current in magnitude. Comparing the two characteristic curves in figure [2], it can be seen that more electrons are emitted from the wire as the heating voltage increases.
To check whether the linear part of the characteristic curves, shown in figure [2], behaves according to the Schottky-Lamuir space charge law [2], the magnitudes of the measured data are logarithmized and plotted against each other. This can be seen in figure [3] and [4]. Linear regression is used to determine the slope of the linear trend.

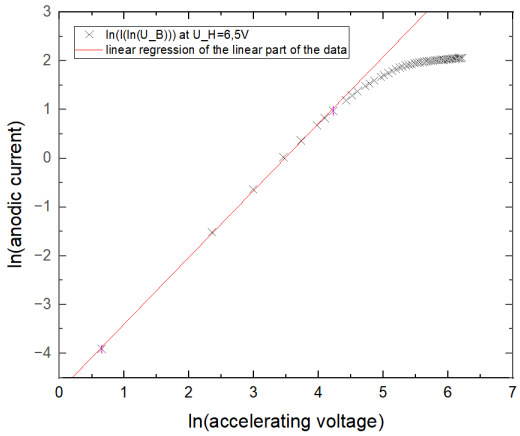

Figure 3: Logarithmic representation of the characteristic curve for $U_H = 6.5V$. The absolute value of the values was used. The slope of the regression line represents the proportionality factor.

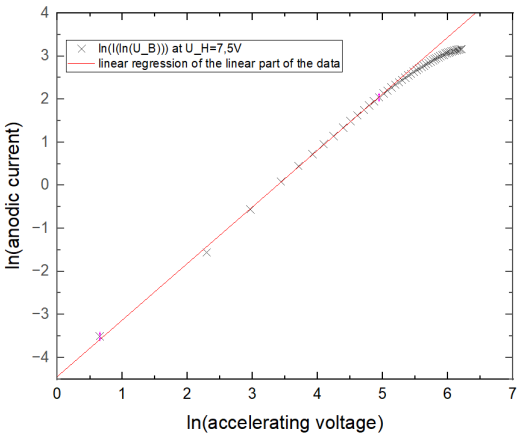

Figure 4: Logarithmic representation of the characteristic curve for $U_H = 7.5V$. The absolute value of the values was used. The slope of the regression line represents the proportionality factor.

Due to the logarithmization of the data, a slope of $m_{U_H} = \frac{3}{2}$. The slope of the regression line in figure [3] is $m_{6.5V} = (1.370 \pm 0.013)$. The slope of the regression line in figure [4] is $m_{7.5V} = (1.313 \pm 0.010)$. The proportionality of the measured values does not match the theoretical proportionality [2]. One possible cause for this could be loss of contact. Another reason could be leftover particles that interact with the electrons.
The contact stresses U_C can be calculated from the determined proportionalities m_{U_H}. It acts like a constant voltage shift in the tube, which means that the externally applied accelerating voltage U_a must be corrected by this amount. Consequently, U_C shifts the entire characteristic curve to the right on the voltage axis, since the theoretical current only begins at $U_a = U_C$. To calculate the contact voltage, the measured values up to $U_a \approx 20V$ are plotted in an graph and a regression line is calculated using linear regression. The data can be seen in the figures [5] and [6].

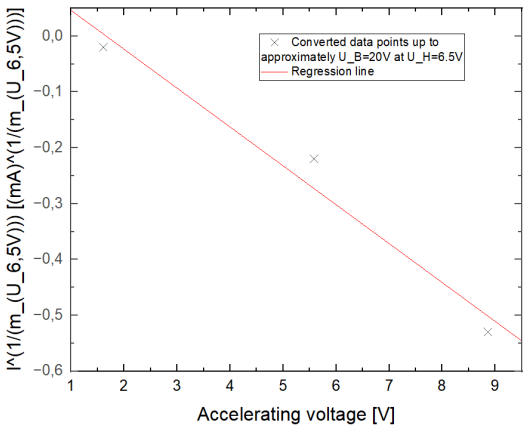

Figure 5: Determination of the contact voltage at $U_H = 6.5V$ using a regression line and the measured data of the characteristic curve up to $U_a \approx 20V$. The data points were converted using equation [**??**].

To find the contact voltage, the values from the linear regression can be used. This is done by expressing the relationship between the anode current I and the accelerating voltage U_B as a linear equation:

$$I^{\frac{1}{m_H}}(U_B) = a \cdot U_B + b \tag{3}$$

Here, m_H is the previously calculated slope, a is the new slope of this line, and b is the crossing of the y-axis.
The contact voltages can be calculated using equation [3] and the values from the linear regression. The calculated contact voltages are $U_{K,6.5V} = (1.7 \pm 1.2)V$ and $U_{K,7.5V} = (2.30 \pm 0.92)V$. The magnitude of the contact voltages is $U_C \approx 10^0 V$.
In the last part of the experiment, the influence of the heating voltage U_H on the saturation current I_S

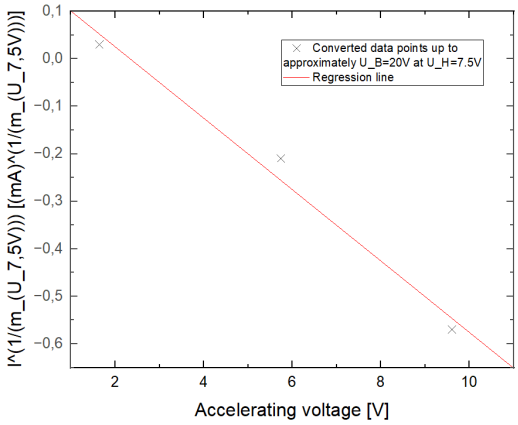

Figure 6: Determination of the contact voltage at $U_H = 7.5V$ using a regression line and the measured data of the characteristic curve up to $U_B \approx 20V$. The data points were converted using equation [**??**].

is investigated. The current density J of the emitted electrons can be described by the Richardson equation [1].
Before analyzing the saturation current, the temperature T of the tungsten wire needs to be determined. This temperature can be calculated from the set heating voltage U_H using the following formula:

$$T(U_H) = 1616\,\mathrm{K} + 97\,\frac{\mathrm{K}}{\mathrm{V}} \cdot U_H \tag{4}$$

The equation [4] can be used to convert the set heating voltage into a temperature.
To determine the saturation current at different heating voltages, an acceleration voltage of $U_a = 500V$ was set. Heating voltages in the range of $U_H = 5.0V$ to $U_H = 7.0V$ were set in $\Delta U_H = 0.1V$ steps. In graph [7] $ln(\frac{I_S}{T^2})$ was plotted against $\frac{1}{T}$. To calculate the work function W_A, the Richardson equation can be rewritten by using the natural logarithm. This results in a linear equation for the saturation current I_S and the temperature T:

$$\ln\left(\frac{I_S}{T^2}\right) = -\frac{W_A}{k_B}\frac{1}{T} + \ln(B) \tag{5}$$

In this equation, W_A represents the work function, k_B is the Boltzmann constant, and B is a material specific constant. By determining the slope of this line, the work function can be easily calculated. The work function can be calculated using equation [5] and the regression line from figure [7].
With the slope $m = (-73.4 \pm 1.1)10^3$, one can calculate a work function of $W_A = (6.322 \pm 0.092)eV$. The literature value for the work function of tungsten is $W_{A,Lit} = 5.3eV$, which is higher than the calculated value. The deviation from the literature value can be caused by residual oxygen atoms in the vacuum. These oxygen atoms can create oxidation

surfaces on the tungsten wire, leading to a higher work function.
Another source of error is the calculated temperature of the tungsten wire, rather than the measured temperature.

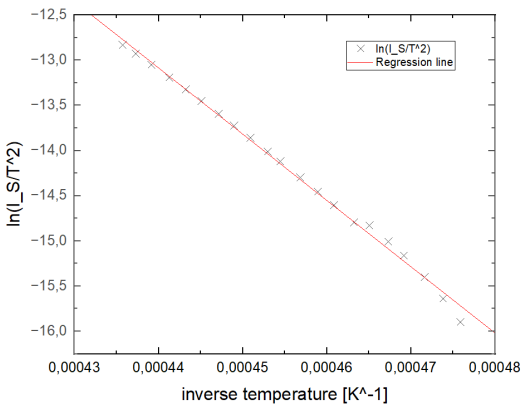

Figure 7: Plotting the logarithmic saturation flow rate against the quadratic temperature as a function of the inverse temperature. The fitted regression line satisfies equation [5], from which the work function W_A can be calculated.

This can lead to a deviation in the work function when the value is substituted into the Richardson equation.

2 Franck-Hertz experiment with neon

2.1 Introduction

The Franck-Hertz experiment was first performed and described by James Franck and Gustav Hertz in 1914. The experiment confirmed Bohr's postulated discrete energy levels of electrons.

In the Franck-Hertz experiment, electrons with a specific kinetic energy are fired at atoms. The kinetic energy is calculated according to equation [6].

$$E_{kin} = e * U_a(t) \tag{6}$$

The electrons bound to the atoms have discrete energy levels. If the kinetic energy of the free electrons is below the difference between two energy levels, an elastic collision occurs. Since the atom is significantly heavier than the electron, the latter retains almost all of its kinetic energy. If the electron has exactly the energy required to excite a bound electron, an inelastic collision occurs. The free electron transfers exactly this energy to the bound electron to excite it. Consequently, the free electron loses energy. If the free electron has more energy than the difference between two energy levels, it excites a bound electron and still retains some kinetic energy. Since the remaining energy is no longer sufficient to excite another electron, the free electron undergoes elastic collisions with atoms. Once a free electron has transferred its energy to a bound electron, it can gain energy again to excite another electron. A bound electron tends to occupy the lowest possible energy state. When an electron is excited, it tends to return to its original state. To do so, the electron emits energy in the form of a photon. The photon has exactly the energy corresponding to the difference between the two energy levels. The energy of a photon is calculated using equation [7].

$$\Delta E = h * \nu \tag{7}$$

2.2 Materials and methods

For this experiment, a glass tube is filled with neon gas at a low gas pressure. On one side, there is a hot cathode K, which emits electrons due to an applied heating voltage U_1. The molecules of the wire ionize and attract the emitted electrons. The electrons repel each other due to the Coulomb force. An equilibrium of forces is established. The grid G_1 is slightly positively charged and attracts the emitted electrons. The voltage must be adjusted so that the electrons have a kinetic energy of $E_{kin} \approx 0\ eV$ when they reach the grid G_1. The electrons gain their kinetic energy through an applied accelerating voltage U_2. The electric field is not constant but increases linearly with time. The resulting force can be described by equation [8].

$$F_{el}(t) = e^{-}\frac{U_a(t)}{d} = e^{-}E(t) \tag{8}$$

The grid G_2 is positively charged and decelerates electrons with low kinetic energy. Between the grids G_1 and G_2, the electrons collide with the neon atoms elastically and, in some cases, inelastically. Electrons with sufficient kinetic energy are collected at the anode. The current can be measured using an ammeter. The experimental setup is shown in figure [8]
For the Franck-Hertz experiment, an evacuated tube is filled with neon gas at low vapor pressure. On one side is a thermionic cathode which is heated by an applied voltage and emits electrons. The molecules of the cathode ionized and attract the released electrons. An electron cloud forms in front of the cathode. This remains stable due to a balance of forces resulting from mutual repulsion and attraction by the cathode. To protect the cathode and better control the energy of the electrons, a weakly positively charged grid G_1 is placed close behind the cathode. The grid pulls the electrons out of the electron cloud and brings them onto the acceleration track. It is important that the attraction by G_1 is

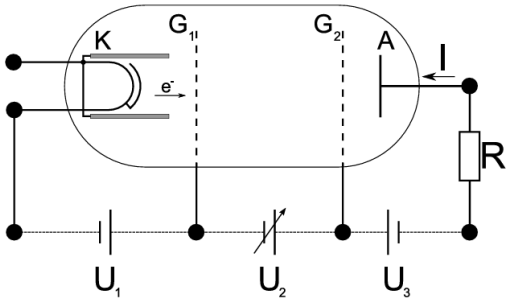

Figure 8: Schematic experimental setup of the Franck-Hertz experiment with neon. $U_1 \hat{=} U_H$, $U_2 \hat{=} U_a$ and $U_3 \hat{=} U_g$. I corresponds to the measured anode current.

minimal, so that the electrons enter the acceleration region with a kinetic energy of $E_{kin} \approx 0eV$. This allows the energy of the electrons to be determined. On the acceleration path, the electrons from G_1 are attracted through the anode.
The voltage at the anode is increased uniformly. The electrons are absorbed at the anode and the current is measured. A grid G_2 is connected in front of the anode, which creates a counter voltage U_g towards the anode. The opposing voltage prevents electrons with low kinetic energy from being included in the current measurement. On the accelerating path between G_1 and G_2, the electrons acquire an increasingly greater kinetic energy due to the increasing electric field.
Because the tube is filled with neon gas, the electrons collide with the neon atoms. When electrons have low kinetic energy, elastic collisions occur, resulting in no energy loss. As the accelerating voltage $U_a(t)$ increases, the current also increases. Given sufficient kinetic energy, electrons and neon atoms collide inelastically. During the collision, the electron transfer a certain amount of energy to the neon atom. The transferred energy brings a valence electron of the neon atom into an excited state. Since atoms, according to Bohr's atomic model, only have dicrete energy levels, electrons can only excite atoms above a certain energy. Even if the electrons have more energy, they can still only release the energy that the atom needs to move from the ground state E_0 to an excited state E_1. The required energy is $\Delta E_{Lit} = 18.9eV$. However, the excited state of the neon atom is not stable. The atom falls back to the ground state E_0 and releases the energy in the form of a photon.
A photon with a wavelength of $\lambda_{Lit} = 585.2nm$ is emitted, which is visible as orange light. With increasing voltage U_a, it is possible that an electron, after an inelastic collision, is accelerated back to the required speed to excite another atom. The emitted electrons excite the atoms at periodic intervals, which is supposed to make the visible light appear separated. The current flow must follow the shape of a linearly increasing sine function.

2.3 Results and discussion

To obtain clearly visible results, the different voltages are varied. Here, $U_H \leq 5V$ and $U_g \leq 10V$ hold true. The accelerating voltage ranges from $0V$ to $80V$.
The maxima of the recorded measurement curve are at $m_1 = 18.73V$, $m_2 = 36.49V$ and $m_3 = 57.29V$, visible in fig. 9.

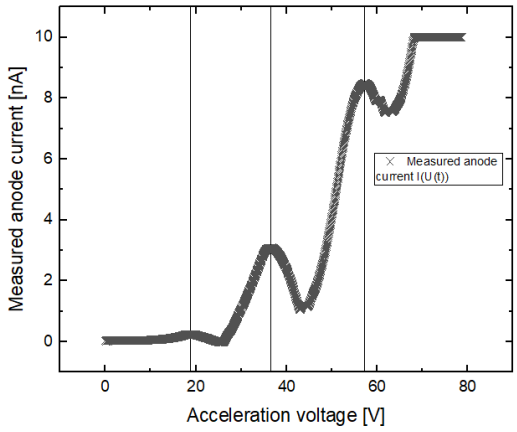

Figure 9: Measured anode current as a function of the accelerating voltage. The heater voltage is $U_H = 0.81V$ and $U_g = 8.57V$.

The values of the maxima have an error of $\sigma_m = \pm 1V$. If one calculates the difference between two maxima and averages the values, one obtains $\Delta U = (19.28 \pm 0.71)V$. The equation [6] can be used to calculate the energy difference of the electrons between to maxima or minima. The anode current has a minimum within a kinetic energy of the electrons of $\Delta E_{kin} = (19.28 \pm 0.71)$. As described in section [2.1], this means that a neon atom needs an energy of $E = (19.28 \pm 0.71)eV$ to reach an excited state. The determined value is consistent with the value in the literature. If the wavelength is calculated using the determined energy via equation [7], one obtains $\lambda = (64.31 \pm 2.4)nm$. The calculated wavelength is not consistent with the orange glow observed during the experiment.
The electron must therefore be transitioned into a higher excited state than the first excited state. The transition to the first excited state is forbidden by the selection rules. From this first possible excited state, the electron drops back into an energetically more favorable excited state. This transition must be allowed by the selection rules. Since the energy difference between these levels is smaller, a photon with a longer wavelength is emitted. The wavelength

of the emitted light is $\lambda_{Lit} = 585.2\,nm$. This results in an energy difference of $\Delta E = 2.11\,eV$ between the higher excited state and the lower excited state. This explains the observed orange light. This suggests that the neon atom was not brought into its first excited state. The neon atom is brought into a higher, unstable state and falls into an excited, stable state, not into the ground state.
During the experiment, it was observed that the luminescence occurs in thin surfaces. The glow occurs above a certain electron energy. As energy increases, luminous planes are added at the same interval of $\Delta U_a = (19.28 \pm 0.71)V$. The distances between the luminous surfaces are equidistant and, with increasing accelerating voltage, move ever closer to the grid G_1. It is visible to the naked eye that the atom can be excited at discrete energy levels.

3 Conclusion

In this experiment, the characteristic curve of a vacuum diode was determined. For this purpose, the anode current was plotted as a function of the accelerating voltage. The study checked whether the proportionality follows the Schottky-Langmuir space charge law, which proved to be false. The determined proportionality constant m_{U_H} between anode current and accelerating voltage is $m_{6.5V} = 1.370 \pm 0.013$ and $m_{7.5V} = 1.313 \pm 0.010$ The contact voltage U_K could be determined from the proportionality between anode current and accelerating voltage. The determined contact voltages are $U_{C,6.5V} = 1.7 \pm 1.2V$ and $U_{C,7.5V} = 2.3 \pm 0.92V$. The influence of the heating voltage on the saturation current was then investigated. These results allowed the work function of the tungsten wire used to be calculated. The work output is $W_A = 6.322 \pm 0.092eV$.
in the second part of the experiment, the Franck-Hertz experiment was carried out with neon. The aim of the experiment was to confirm the excitability of atoms in discrete energy levels. This prediction is made by Bohr's atomic model. For this purpose, the positions of the maxima were determined and their distances were calculated. The electron, with a kinetic energy of $E_{kin} = (19.28 \pm 0.71)V$ transfers energy to a neon atom, which excites it. The transfer of energy from an electron to an atom is visible as a minimum. The emitted wavelength could be calculated, the result does not match the observed orange light. This suggests that the neon atom reaches a higher excited state than the first excited state and does not fall directly back to the ground state.

Sources

References

[1] Meschede, D., Gerthsen Physik (25th ed.), Springer Spektrum Verlag, 2015.

[2] Haken & Wolf, Atom- und Quantenphysik (8th ed.), Springer-Lehrbuch Verlag, 2004.

[3] Demtröder, W., Experimentalphysik 3 (5th ed.), Springer Spektrum Verlag, 2016.